Reproducing Decision Transformer and Improving Its Trajectory Stitching with Expectile-Based Return Relabeling

EE AI for Robotics, 2026 Fall | Final Project Proposal | **Track 1**

1. Problem Description

Decision Transformer (DT; Chen et al., NeurIPS 2021) casts reinforcement learning (RL) as conditional sequence modeling: a causally masked Transformer, conditioned on the desired return, past states, and actions, outputs actions. It replaces dynamic-programming bootstrapping, which is unstable with function approximation, by a supervised sequence-modeling loss. However, DT lacks *stitching* ability, i.e., learning a good policy by combining pieces of sub-optimal trajectories, which matters especially when the offline dataset contains only sub-optimal trajectories. This problem is important to me because it exposes a fundamental trade-off between supervised and value-based learning from fixed data, a central theme of this course. The question is especially relevant to robotics: real robots cannot afford large numbers of trial-and-error episodes, so learning good controllers from previously logged, mixed-quality data is highly valuable. Following Track 1, I will (a) reproduce DT on standard offline RL benchmarks with simulated robots and (b) test an improvement that injects value information into DT's return conditioning.

2. Relevant Work

- **Decision Transformer** (Chen et al., NeurIPS 2021): the paper to be reproduced.
- **Implicit Q-Learning, IQL** (Kostrikov et al., ICLR 2022): offline RL that never queries the value of actions unseen in the data, using an upper expectile of the value function; strong results on D4RL. It serves as a value-based baseline and as the source of the value function in my improvement.
- **Q-learning Decision Transformer, QDT** (Yamagata et al., ICML 2023): uses dynamic-programming (Q-learning) results to relabel the return-to-go in the training data, then trains DT on the relabeled data to gain stitching ability. It is the closest related work.
- **D4RL** (Fu et al., 2020): standard benchmark suite of offline RL datasets of varied quality.

3. Research Plan

Phase 1: Reproduction. Implement DT (or adapt the official code) and evaluate on D4RL MuJoCo tasks (HalfCheetah, Hopper, Walker2d; medium, medium-replay, medium-expert) over several seeds. Compare normalized scores with the published ones and document any discrepancy.

Phase 2: Improvement. Train an IQL value function on each dataset and use it to relabel the return-to-go before training DT (an IQL-based variant of the QDT idea, whereas QDT relies on Q-learning). Compare DT, QDT-style relabeling, IQL, and my variant, including a stitching-focused setting on sub-optimal data where DT is expected to struggle. To move beyond locomotion, I will also evaluate on Franka Kitchen, a D4RL robot-manipulation task in which a robot arm must complete a sequence of kitchen subtasks, a natural setting for stitching. The idea may or may not work; a failure analysis will be part of the report.

Phase 3: Analysis and report. Ablations on context length, relabeling strength, and dataset quality. Metric: normalized D4RL score (mean and standard deviation over seeds).

Expected outcomes. (i) A verified reproduction of DT; (ii) a controlled comparison of sequence-modeling and value-based offline RL; (iii) an assessment of whether expectile-based relabeling improves stitching; (iv) a publication-style final report and presentation.

References. Chen et al., Decision Transformer: Reinforcement Learning via Sequence Modeling, NeurIPS 2021. Kostrikov, Nair, Levine, Offline Reinforcement Learning with Implicit Q-Learning, ICLR 2022 (arXiv:2110.06169). Yamagata, Khalil, Santos-Rodriguez, Q-learning Decision Transformer, ICML 2023 (PMLR 202:38989-39007). Fu, Kumar, Nachum, Tucker, Levine, D4RL: Datasets for Deep Data-Driven Reinforcement Learning, 2020 (arXiv:2004.07219).